Introduction to Embedded Computer Systems

ECE 3430 – Fall 2026

Charles L. Brown Department of Electrical and Computer Engineering — University of Virginia

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Teaching Assistants	Kevin Li, agp9dv@virginia.edu Maximilian Meer, qqx3uf@virginia.edu Seyed Hamid Reza Roodabeh, ydq9ag@virginia.edu
Class Meetings	**Mondays and Wednesdays, 11:00 AM – 12:50 PM** Mechanical Engineering Building, Room 213
Instructor Office Hours	Mondays/Wednesdays, 1:00–2:00 PM, in person Zoom or other times by appointment
TA Office Hours	In person in the Link Lab Arena: Tuesdays: TBD • Wednesdays: TBD • Thursdays: TBD • Fridays: TBD Zoom or other times; request by email.
Course Websites	**UVA Canvas:** https://canvas.its.virginia.edu/courses/189740 **Piazza:** https://piazza.com/virginia/fall2026/ece3430001alemzadeh/home **Public Page:** https://homa-alem.github.io/teaching/ece_3430/index.html
Primary Textbook	***Introduction to Embedded Systems Using the MSPM0+,* Jonathan W. Valvano,** ISBN: 979-8852536594, July 2023. **- Free e-book:** https://users.ece.utexas.edu/~valvano/mspm0/ebook/index.htm - Print copy on Amazon: https://www.amazon.com/Introduction-Embedded-Systems-Using-MSPM0/dp/B0C9SB2QQ9
Required Hardware, Software, and Tools	- **A personal laptop** with a USB port to install and run CCS. - **Code Composer Studio (CCS)** IDE by Texas Instruments — free download. - **TI LP-MSPM0G3507** LaunchPad development board (ARM Cortex-M0+) - **Educational BoosterPack MKII** (LCD, buzzer, joystick, and sensors) - **Adafruit STEMMA Speaker** and Plug and Play Audio Amplifier Loaner development boards and devices are provided by the ECE department: - Each student will borrow a TI LaunchPad, an Educational BoosterPack MKII, and a STEMMA Speaker and must return them at the end of the semester.
Additional References	- LP-MSPM0G3507 datasheet and technical reference manual - LaunchPad and BoosterPack user guides and schematics - MSPM0 software development kit (SDK) - MSPM0+ Assembly Instruction Set and reference manual

Course Description

Embedded systems are special-purpose computers at the core of Cyber-Physical Systems (CPS) that monitor and control physical processes through real-time interactions with sensors and actuators. More than 90% of all manufactured microprocessors are embedded in everyday devices — from airplanes, automobiles, and medical devices to digital cameras, home appliances, smart buildings, IoT devices, and robots. What are the building blocks of an embedded system? How does software running on a microcontroller observe and act on the physical world? How can we design an embedded system and verify that it satisfies specific functional and timing requirements?

This lab-based course will help you answer these questions by providing the foundational knowledge and hands-on experience needed to design, evaluate, and implement embedded computer systems, with an emphasis on the architecture and interface issues at the boundary between hardware and software. You will develop software in C and ARM assembly that executes on a microcontroller-based development board, interfacing directly with physical inputs and outputs. Topics include:

- **Embedded system architectures:** ARM Cortex-M0+ microcontrollers (TI MSPM0G3507); ARM assembly programming.
- **Embedded software development:** Embedded C programming and toolchains; modular design, debugging, and profiling; finite state machines (FSMs).
- **Embedded input/output (I/O):** Memory-mapped I/O; I/O programming; I/O synchronization and interrupts; hardware timers; pulse-width modulation (PWM); serial communication interfaces (UART, SPI, I^2C); analog interfacing (ADC, DAC).

Pre-requisites

Must have completed (ECE 2300 or ECE 2630) AND ECE 2330 and CS 2130, all with C- or better. A working knowledge of digital logic, computer organization, and C programming is assumed.

Learning Objectives

This course will provide the foundational knowledge and technical skills to design, implement, and evaluate embedded computer systems. Upon successful completion of the course, you will be able to:

1. Describe the key building blocks of embedded system architectures and their function, including the ARM Cortex-M0+ CPU, memory, memory-mapped I/O, I/O synchronization, and interrupts.
2. Write, compile, and debug programs in C and ARM assembly language in an embedded programming environment (Code Composer Studio) for a microcontroller-based development board.
3. Develop embedded software that interfaces with physical inputs and outputs using digital I/O, interrupts, timers, PWM, serial communication (UART, SPI, I^2C), and analog conversion (ADC, DAC).
4. Apply embedded software design practices and tools (modular design, data flow graphs, finite-state machines) to design an embedded information processing system according to a given set of requirements, in collaboration with your team members.
5. Verify that your design and implementation satisfy all given functional and timing requirements (confirming that you *designed the correct system* and that you *designed the system correctly*) using the IDE's debugging features and laboratory equipment.
6. Direct your own learning using available technical resources, such as datasheets, reference manuals, and schematics.
7. Clearly and concisely communicate your design and verification results to a professional standard.

Activities, Assessments, and Grading

Your final grade comprises the following components, which reinforce and assess the learning objectives. For each component, the table below summarizes how much it counts, how you may collaborate, and what you must submit and demonstrate to receive credit.

Component	Weight	Collaboration	Submission & Demo
In-Class Activities	**30%**	Peer collaboration/discussion allowed. **Quizzes:** No help is allowed; individual, closed-book submissions.	Individual Canvas submission and/or in-person demo to TAs or instructor. *No late submissions allowed.*
Learning Activities	**20%**	Peer collaboration/discussion allowed. Individual take-home implementation.	Individual Canvas submission and in-person demo to TAs or instructor.
Assessments	**30%**	No peer collaboration or discussion and no TA help allowed. Individual take-home implementation.	Individual in-class design, Canvas submission, and in-person demo.
Final Project	**20%**	Team collaboration required. Every member must contribute to code, demo, and own their part.	Team Canvas submission, in-person demo and final presentation. *No late submissions allowed.*

In-Class Activities (30%): In-class activities are guided, hands-on work completed during the class period, often by following prescribed instructions, and include interfacing with real hardware on the LaunchPad, reading datasheets, short, timed Canvas quizzes, and applying concepts from lectures. You may use any available resources (subject to the ***AI Usage Policy***) and work together, but any code you demonstrate must ultimately be authored and verified by you. To receive credit, you demonstrate your working solution to a TA or instructor in person. During your demonstration, you may be asked to explain your code or make a minor, live modification to it to confirm your understanding (see the ***Ownership Rule***). Points for verified solutions are awarded "all or nothing" (effectively a 1 or 0). In-class activities are typically designed to be completed and checked off during the class period; if you actively participate in class but need more time to finish debugging, you may check off at the following session or during office hours without penalty. Students who cannot attend class due to illness or a personal or professional absence (see the ***Absences, Illnesses, and Career Development*** policy) may, *with the instructor's permission*, complete the activity individually and demonstrate it at the following session.

Learning Activities (20%): Learning activities offer more independent, experiential opportunities to apply what you did in class, with TAs and instructors assisting as needed. You may use any available resources (subject to the ***AI Usage Policy***) and may work together, but any code you submit must ultimately be code that you created and verified, and you must be able to explain it. In addition to in-person verification, you must submit all the required submission material specified in the assignment instructions on Canvas to receive full credit. Points for verified solutions are awarded "all or nothing" (effectively a 1 or 0).

Assessments (30%): To assess your ability to design solutions for comprehensive, "real-world" problems, three pledged, individual take-home exercises will be administered during the semester. Each assessment is introduced in class with an associated design activity, which you then complete individually as a take-home exercise. All assessments are open-book and open-notes, but limited to the resources available to you through the course website. You may NOT work together; you may NOT consult current or former students or TAs; and you may NOT use any generative AI tools, even for assistance. You will submit a complete project archive (zip) on Canvas for testing and verification, and you will demonstrate your working solution in person during one of the class sessions following the due date. Violation of any of these rules will result in a zero on the assessment and referral to the Honor Committee.

Final Project (20%): Building on the knowledge and skills learned through lectures, class activities, and assessments, you will work in teams of 2 students on designing an embedded system on the LaunchPad and the Educational BoosterPack MKII. Teams sign up and are finalized upon project introduction. The project is structured as a set of leveled requirements (Levels 0–2, plus bonus features), with two graded progress checks, a final project submission in the last week of classes, and an in-person demo and presentation during the final exam period. Projects will be evaluated based on the application of concepts learned in class, the demonstration of a working prototype at each level, and the quality of the final demo and presentation. All team members are expected to participate in the final demo/presentation.

Course Schedule

The course schedule outlines the weekly topics, activities, and due dates throughout the semester. This schedule is tentative and subject to change depending on our progress. I will inform you of any updates or changes at the beginning of each class, on Canvas, and on Piazza. I expect everyone to check the schedule regularly on Canvas or on the public course page before each class: https://homa-alem.github.io/teaching/ece_3430/schedule.html

Online Tools and Resources

Class meetings (lectures and in-class activities) will be in person. All students are expected to attend class **in person**. To support asynchronous learning, the lectures will be recorded via **Zoom**; there is no remote attendance option (see the ***Recording of Classroom Activities*** policy). The lecture recordings, slides, reading materials, and all assignments will be posted on **UVA Canvas**, where you will also submit your work. We will use **Piazza** to post announcements, run polls, and host Q&A. You are expected to actively follow announcements and posts on Canvas and Piazza and participate by posting questions, answering classmates' questions, and taking part in polls and online surveys in a timely manner.

Course Policies

Late Submissions: In-class activities and final project deliverables cannot be submitted late. Deadlines for in-class activities are set after the class session, so that if you need more time to finish, you can check off at the following session or during office hours (see In-Class Activities above). Your lowest in-class activity grade will be dropped.

For take-home learning activities and assessments, you have a grace period of three late school days across the entire course, to accommodate unexpected events, such as illness, travel, other deadlines, or interviews, without penalty. Once your grace days are exhausted, late submissions incur a 10% penalty per school day (weekend days do not count). Submissions more than 5 school days late are not accepted, except in cases of extenuating circumstances approved by the instructor. Late days are totaled and penalties applied at the end of the semester.

Regrades: If you believe your work was graded incorrectly, you may request a regrade by writing to the instructor within one week after grades are released, with a clear explanation of the suspected error. Regraded work is evaluated from scratch, and the resulting grade may be higher or lower than the original.

Absences, Illnesses, and Career Development: It is in the best interest of everyone in our community to minimize the spread of infectious diseases. If you are sick with a communicable illness, please stay home, catch up by watching the posted lecture recording, and let me know by email as soon as possible if you are going to miss a class or in-class activity. Individual completion of in-class activities is allowed only with *prior instructor approval via email*. The same policy applies when you need to miss class for career development, such as interviews, conferences, or other events that support your growth as an engineering professional. Please notify me by email as far in advance as possible to arrange appropriate accommodations.

Academic Integrity: I expect every student in this course to fully comply with all the provisions of the University's Honor Code. By enrolling in this course, you have agreed to abide by and uphold the Honor System of the University of Virginia (https://honor.virginia.edu), as well as the following specific policies:

1. All graded assignments must be pledged where indicated.
2. For all assessments, you must sign the University of Virginia Honor Pledge to indicate that:
   - You have not used any unauthorized resources. Unauthorized resources include other current students, former students, teaching assistants, other faculty, and generative AI. In addition, I will explicitly announce any restrictions on access to materials (including the textbook, materials on the course website, and materials available on websites external to the course website). In general, for an assessment, I am the only authorized resource for assistance unless otherwise explicitly stated by me.
   - You abided by any time limitations or constraints for the assessment.
   - You solely performed any work as relates to the assessment.
3. For learning activities, you are encouraged to collaborate with other students. However, regarding your submission, you are required to sign the University of Virginia Honor Pledge to indicate that you have not used any unauthorized resources and that what you submit was solely created and verified by you, unless otherwise explicitly stated by me (for example, for a team-oriented learning activity where team submissions are permitted).
4. Except for assessments, you should feel free to consult with teaching assistants (both undergraduate and graduate) and use material available to you on the course website, as well as material accessible to you on websites external to the course website. However, any attempt to take credit for work done by another person is considered plagiarism and a violation of the Honor Code. You must properly credit any team members with whom you collaborate; and **you must cite any resources or individuals you consult or any code or tools that you use.** You should always understand the material or code you use and be able to explain it.
5. All suspected violations will be forwarded to the Honor Committee, and you may, at my discretion, receive an immediate zero on that assignment.

Please let me know if you have any questions about the course Honor policy. If you believe you may have committed an Honor offense, you may wish to file a Conscientious Retraction by calling the Honor offices at (434) 924-7602. For your retraction to be considered valid, it must, among other things, be filed with the Honor Committee before you are aware that the act in question has come under suspicion by anyone.

AI Usage Policy: Generative artificial intelligence tools are now part of professional embedded systems engineering, and you will be expected to use them competently in your future career. They are also demonstrably unreliable and can prevent you from learning the very skills that make you employable. This policy governs all such tools, including the AI features built into Code Composer Studio. Violations of this policy are handled under the University's Honor System. In this course, every assignment is labeled for AI usage, indicating whether **"No AI"** or **"AI Support"** is allowed:

- **No AI:** You may **not use generative AI on assessments, quizzes, design activities**, or the ARM assembly activities in Module 1. This foundational material must live in your head. You may still use the textbook, datasheets, lecture materials, and human help per the collaboration policy.
- **AI Support:** For everything else, you may use AI to support your work (see the allowed uses below), but you **may NOT use it to generate the core logic of the assignment** (the parts the assignment exists to teach), which is identified in each handout. Whenever you use AI for an allowed purpose, **cite the tool and briefly describe how you used it**.

You **may use AI tools** (in AI Support assignments and for studying and preparation) to:

- Look up technical concepts and terminology, ask for explanations and examples (e.g., how a peripheral register or a C construct works), and resolve conceptual ambiguities
- Explain compiler errors, linker errors, and fault conditions, and debug syntax errors
- Answer syntax questions and provide general C language help, including learning about library functions
- Generate initialization boilerplate and project scaffolding
- Check the grammar and readability of your writing; make plots and figures; format print statements
- Write practice problems to prepare for the quizzes and assessments

You **<u>may NOT use AI tools</u>** on any assignment to:

- Write your solutions, including activity or assessment code, the core logic of any assignment, or your project's functional code
- Produce code or text that you copy into your work without understanding it. You must be able to explain every line (the Ownership Rule)
- Complete assignments without learning the underlying concepts
- Do any part of work labeled No AI.
- Generate the text of your reports or the slides of your final project presentation

The Ownership Rule: You own every line of code you submit, regardless of where it came from. It is your responsibility to ensure the quality, integrity, and accuracy of everything you submit. At any check-in, you may be asked to explain any line and to modify it on the spot. An inability to do so will be treated as evidence that the work is not yours, and the check-in will not be credited.

Recording of Classroom Activities: Lectures will be recorded to support review and students who must miss class, and the recordings will be posted on UVA Canvas. Because these recordings may visually or audibly identify you and/or your fellow students, or contain sufficient context to identify a student, they may only be used for individual or group study with the other students enrolled in this class during this semester. They may be stored only on University-owned, password-protected sites, such as UVA Canvas. You may not distribute them, in whole or in part, through any other platform or to any person outside of this class, nor may you make your own recordings of this class unless written permission has been obtained from me and all participants in the class have been informed that recording will occur. For additional details, see Provost Policy 005 (https://uvapolicy.virginia.edu/policy/PROV-005).

Students with Disabilities or Learning Needs: It is my goal to create a learning experience that is as accessible as possible. If you anticipate any issues related to the format, materials, or requirements of this course, please meet with me outside of class so we can explore potential options. Students with disabilities may also wish to work with the Student Disability Access Center (SDAC) to discuss a range of options to removing barriers in this course, including official accommodations. We are fortunate to have an SDAC advisor, Courtney MacMasters, physically located in Engineering. You may email her at cmacmasters@virginia.edu to schedule an appointment. For general questions, please visit the SDAC website: sdac.studenthealth.virginia.edu. If you need any special assistance, please get in touch with me as soon as possible by email or during office hours. If you have already been approved for accommodations through SDAC, please send me your accommodation letter and meet with me so we can develop an implementation plan together.

Harassment, Discrimination, and Interpersonal Violence: The University of Virginia is dedicated to providing a safe and equitable learning environment for all students. If you or someone you know has been affected by power-based personal violence, more information can be found on the UVA Sexual

Violence website, which describes reporting options and resources available at www.virginia.edu/sexualviolence. The same resources and options for individuals who experience sexual misconduct are available for discrimination, harassment, and retaliation. UVA prohibits discrimination and harassment based on age, color, disability, family medical or genetic information, gender identity or expression, marital status, military status, national or ethnic origin, political affiliation, pregnancy (including childbirth and related conditions), race, religion, sex, sexual orientation, or veteran status. UVA policy also prohibits retaliation for reporting such behavior.

If you witness or are aware of someone who has experienced prohibited conduct, you are encouraged to submit a report to Just Report It (justreportit.virginia.edu) or contact EOCR, the Office of Equal Opportunity and Civil Rights. If you would prefer to disclose such conduct to a confidential resource where what you share is not reported to the University, you can turn to Counseling & Psychological Services ("CAPS") and Women's Center Counseling Staff and Confidential Advocates (for all students).

As your professor and as a person, know that I care about you and your well-being and stand ready to provide support and resources as I can. As a faculty member, I am a responsible employee, which means that I am required by University policy and by federal law to report certain kinds of conduct that you report to me to the University's Title IX Coordinator. The Title IX Coordinator's job is to ensure that the reporting student receives the resources and support that they need, while also determining whether further action is necessary to ensure survivor safety and the safety of the University community.

Religious Accommodations: It is the University's long-standing policy and practice to reasonably accommodate students so that they do not experience an adverse academic consequence when sincerely held religious beliefs or observances conflict with academic requirements. If you wish to request academic accommodation for a religious observance, please submit your request directly to me by email as far in advance as possible. Students who have questions or concerns about academic accommodations for religious observance or religious beliefs may contact the University's Office for Equal Opportunity and Civil Rights (EOCR) at UVAEOCR@virginia.edu or 434-924-3200.

Support for Your Career Development: Engaging in your career development is an important part of your student experience. For example, presenting at a research conference, attending an interview for a job or internship, or participating in an externship or shadowing experience are not only necessary steps on your path but also invaluable lessons in and of themselves. I wish to encourage and support you in your career development activities. To that end, please notify me by email as far in advance as possible to arrange for appropriate accommodations.

Student Support Team: You have many resources available to you when you experience academic or personal stress. In addition to your professor, the School of Engineering and Applied Science has staff members located in Thornton Hall who you can contact to help manage academic or personal challenges. Please do not wait until the end of the semester to ask for help!

Learning

- Lisa Lampe, Assistant Dean for Undergraduate Affairs
- Georgina Nembhard, Director of Undergraduate Success
- Courtney MacMasters, Accessibility Specialist
- Free tutoring is available for most classes.

Health and Wellbeing

- Kelly Garrett, Assistant Dean of Students, Student Safety and Support

- Elizabeth Ramirez-Weaver, CAPS counselor*
- Katie Fowler, CAPS counselor*

*You may schedule time with the CAPS counselors through Student Health (studenthealth.virginia.edu/mental-health/getting-started-caps). When scheduling, be sure to specify that you are an Engineering student. You are also urged to use TimelyCare (timelycare.com/uva) for either scheduled or on-demand 24/7 mental health care.

Community and Identity

The Center for Connection (The Connect) is a dedicated student space within UVA Engineering that fosters academic success and personal growth. Through its programs and initiatives, The Connect helps students strengthen their engineering identity while providing resources to help them thrive during their studies and beyond. The Connect features an open study area, a flexible event space, and on-site staff who provide direct support and advising to students. It is part of the Office of Community, Opportunity, and Engagement.